

Personal Information



Spanish 07/09/1991

Darías Holgado

Senior Postdoc

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Summary

I began my career as a researcher in 2013 and, thanks to securing external funding, have been working at the university almost continuously ever since. I secured research funding as early as my bachelor's and master's studies, which continued throughout my doctoral program. With more than 10 years of experience, I have conducted multiple data collection projects, including several clinical trials, performed numerous advanced analyses using Matlab and R Studio, and published the results in international journals. I have also conducted several meta-analyses and have experience supervising master's theses.

Keywords: Subjective experience; Meta-science; Experimental psychology; Exercise performance; tDCS; Cognition

1. Work experience

Date	Position and Responsibilities
08/2025 now	Senior Postdoc, FernUni Schweiz. Schweiz • Lecturer M04: Research Methods and Statistics II • Data Collection in Experimental Studies • Data Analysis with MATLAB and R • Conducting Meta-Analyses and Systematic Reviews • Publication and Communication of Scientific Results • Applying for external Funding and Writing Research Proposals • Supervising bachelor's and master's Theses and Research Projects
01/2024 07/2025	Research fellow at the Institute of Sports Sciences at the University of Lausanne, Switzerland, and Rey Juan Carlos I University, Madrid, Spain. • Data collection in experimental studies • Data analysis using MATLAB and R • Conducting meta-analyses and systematic reviews • Publication and communication of scientific findings • Securing external funding and drafting research proposals • Supervising bachelor's and master's theses as well as research projects

01/2022 12/2024	Postdoctoral researcher at the Institute of Sports Sciences at the University of Lausanne, Switzerland, and at the Center for Research on Mind, Brain, and Behavior at the University of Granada, Spain • Teaching • Data collection in experimental studies • Data analysis using MATLAB and R • Management of research projects • Conducting meta-analyses and systematic reviews • Publication and communication of scientific results • Acquisition of external funding and drafting of research proposals • Supervision of master's theses and research projects • Participation in clinical trials in accordance with the principles of good clinical practice
04/2021 10/2021	Intern as a research assistant at the Swiss Cycling Federation. • Scientific Advisor • Data collection for the field trial for the 2020 Tokyo Olympics
10/2019 04/2020	Postdoctoral researcher at the Center for Mind, Brain, and Behavior at the University of Granada, Spain • Conducting clinical trials in accordance with good clinical practice • Conducting meta-analyses and systematic reviews • Leading research projects • Writing and communicating scientific findings • Drafting project proposals and securing external funding • Supervising undergraduate students • Data analysis using MATLAB and R Studio
09/2018 12/2018	Ph.D. internship at Campus Biotech, University of Geneva, Switzerland. • Writing and communicating scientific findings • Data analysis using MATLAB and R Studio
10/2015 09/2019	Ph.D. student at the Institute of Sports Science and the Research Center for Mind, Brain, and Behavior at the University of Granada, Spain. • Conducting clinical trials in accordance with the principles of good clinical practice • Leading research projects • Data collection in experimental studies • Data analysis using MATLAB and R Studio • Conducting meta-analyses and systematic reviews • Writing and communicating scientific findings • Drafting project proposals and securing external funding • Supervising undergraduate students • Teaching university courses
07/2015 10/2015	Research Associate at the Faculty of Sport and Exercise Sciences at the University of Kent, United Kingdom • Data collection in experimental studies • Data analysis • Writing and communicating scientific findings
04/2014 10/2015	Research Assistant at the Institute of Sports Science at the University of Granada, Spain • Data collection in experimental studies • Data analysis • Writing and communicating scientific findings • Writing project proposals and securing external funding

09/2013 08/2014	Research Assistant at the Institute of Sports Science at the University of Granada, Spain • Data collection in experimental studies • Data analysis • Writing and communicating scientific findings
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1.2 Education

September 2019	International Doctorate in Biomedicine ¹ . University of Granada. Spain. “Executive functions, self-paced exercise and cycling performance”. Supervisors: Daniel Sanabria and Mikel Zabala. <i>Thesis Common.</i> https://osf.io/preprints/thesiscommons/vtyfu
Oktober 2015	Master of Research in Sport und Physical Activity. University of Granada. Spain
Juli 2014	BSc. in Sport Science (Extraordinary price). University of Granada. Spain

2.1 Major Research Achievements

5 most relevant publications

1. **Darías, Holgado**, Alice Cailleux, Paolo Ruggeri, Corinna Martarelli, Tristan Bekinschtein, Daniel Sanabria, and Nicolas Place. (2025) « Individualized Cognitive Effort to Failure Does Not Affect Subsequent Strenuous Physical Performance». *Medicine & Science in Sports & Exercise* 57(7): p 1603-1615, doi : 10.1249/MSS.0000000000003669.
2. **Darías Holgado**, Daniel Sanabria, Miguel A. Vadillo and Rafael Román-Caballero. (2024). Zapping the brain to enhance sport performance? Evidence from an umbrella review of the effect of transcranial direct current stimulation on physical performance. *Neuroscience & Biobehavioral Reviews*.
<https://www.sciencedirect.com/science/article/pii/S0149763424002902?via%3Dihub>
3. Luis F. Ciria, Rafael Román-Caballero, Miguel Vadillo, **Darías Holgado**, Antonio Luque- Casado, Pandelis Perakakis and Daniel Sanabria. (2023). An umbrella review of randomized control trials on the effects of physical exercise on cognition. *Nature Human Behavior*.
<https://www.nature.com/articles/s41562-023-01554-4>
4. **Darías Holgado**, Cristian Mesquida and Rafael Román-Caballero. (2023). Assessing the evidential value of mental fatigue and exercise research. *Sports Medicine*.<https://rdcu.be/dlHa9>
5. **Darías Holgado**; Thomas Zandonai; Mikel Zabala; James Hopker; Pandelis Perakakis; Antonio Luque Casado; Luis Ciria; Eduardo Guerra Hernandez and Daniel Sanabria. (2017). Tramadol effects on physical performance and sustained attention during a 20-min indoor cycling time-trial: A randomised controlled trial. *Journal of Science and Medicine in Sport*. 01/11/2017.
<https://doi.org/10.1016/j.jsams.2017.10.032>

Liste der Publikationen

¹ Bitte beachten Sie, dass es sich bei dem Namen lediglich um die offizielle Bezeichnung des Programms handelt, der Inhalt und Forschungsschwerpunkt jedoch im Bereich der Psychologie lag.

6. **Darías, Holgado**, Ludovic Leubaz, Paolo Ruggeri, Guillermo Borrigan, Tristan Bekinschtein, Daniel Sanabria, and Nicolas Place (2026). Cognitive endurance after intense physical effort. *Behavioural Brain Research*, 497, 115874. <https://doi.org/10.1016/j.bbr.2025.115874>
7. Lukas Röseler, et al. (2024). The Replication Database: Documenting the Replicability of Psychological Science. *Journal of Open Psychology Data* <https://openpsychologydata.metajnl.com/articles/10.5334/jopd.101>
8. **Darías Holgado**, Léo Jolidon, Guillermo Borrigan, Daniel Sanabria and Nicolas Place. (2023). Individualized mental fatigue does not impact neuromuscular function and exercise performance. *Medicine and Science in Sport and Exercise*. https://journals.lww.com/acsm-msse/Fulltext/9900/Individualized_Mental_Fatigue_Does_Not_Impact.292.aspx
9. Thomas Zandonai, **Darías Holgado**, Luis F. Ciria, James Hopker, Mikel Zabala, Tristan Bekinschtein, and Daniel Sanabria Sanabria, D. (2021). Novel evidence on the effect of tramadol on self-paced high-intensity cycling. *Journal of Sport Science*. Doi: 10.1080/02640414.2021.1877440
10. **Darías Holgado**; Esther Troya; José C. Perales; Miguel A. Vadillo and Daniel Sanabria. (2020). "Does mental fatigue impair physical performance? A replication study" *European Journal of Sport Science*. <https://doi.org/10.1080/17461391.2020.1781265>.
11. **Darías Holgado**; Daniel Sanabria; José C. Perales and Miguel A. Vadillo. (2020). "Mental fatigue might be not so bad for exercise performance after all: A Systematic Review and Bias-sensitive Meta-analysis." *Journal of Cognition*. <https://doi.org/10.5334/joc.126>.
12. **Darías Holgado**; Mikel Zabala and Daniel Sanabria. (2019). No evidence of the effect of cognitive load on self-paced cycling performance. *Plos One*. <https://doi.org/10.1371/journal.pone.0217825>
13. **Darías Holgado**; Thomas Zandonai; Luis F Ciria; Mikel Zabala; James Hopker and Daniel Sanabria. (2019). Transcranial direct current stimulation (tDCS) over the left prefrontal cortex does not affect time-trial self-paced cycling performance: Evidence from oscillatory brain activity and power output. *Plos One*. <https://doi.org/10.1371/journal.pone.0210873>
14. **Darías Holgado**; Miguel A. Vadillo and Daniel Sanabria. (2018). The effects of transcranial direct current stimulation on objective and subjective measures of sports performance: A systematic review and meta-analysis. *Brain Stimulation*. <https://doi.org/10.1016/j.brs.2018.12.002>.
15. Ana Peinado; **Darías Holgado**; Antonio Luque Casado; Miguel Rojo Tirado; Daniel Sanabria Lucena; Coral González; Manuel Mateo-March; Cristobal Sanchez Muñoz; Francisco Calderón; Mikel Zabala. (2018). Effect of induced alkalosis on performance during a field-simulated BMX cycling competition. *Journal of Science and Medicine in Sport*. <https://doi.org/10.1016/j.jsams.2018.08.010>

16. Luis F Ciria; Antonio Luque-Casado; Daniel Sanabria; **Darías Holgado**; Plamen Ch Ivanov and Pandelis Perakakis. (2018). Oscillatory brain activity during acute exercise: Tonic and transient neural response to an Oddball task. *Psychophysiology*. 10.1111/psyp.13326

Reviews and Comments

17. **Darías, Holgado**, Daniel Sanabria, and Nicolas Place (2026). Do all paths lead to mental fatigue or is it merely a self-fulfilling prophecy? *Sports Medicine*. 10.1007/s40279-026-02400-z.
18. Ciria, Luis F., Rafael Román-Caballero, Miguel A. Vadillo, **Darias Holgado**, Antonio Luque-Casado, Pandelis Perakakis, and Daniel Sanabria. (2024) « Reply to: Do Not Underestimate the Cognitive Benefits of Exercise ». *Nature Human Behaviour*, 1-3. <https://doi.org/10.1038/s41562-024-01950-4>.
19. **Darías Holgado**, Agustín Manresa-Rocamora, Lorenzo Zamboni, Fabio Lugoboni, Ana Peiró, and Thomas Zandonai (2022). The effect of benzodiazepines on exercise in healthy adult participants: A systematic review. *Journal of Addictive Diseases*, 0(0), 1–9. <https://doi.org/10.1080/10550887.2021.1990640>
20. **Darías Holgado** and Thomas Zandonai. (2022) Comment on “What is the Effect of Paracetamol (Acetaminophen) Ingestion on Exercise Performance? Current Findings and Future Research Directions.” *Sports Med*. 2022; 52:2561–2.
21. **Darias Holgado** and Daniel Sanabria. (2020). “Does self-paced exercise depend on executive processing? A narrative review of the current evidence.” *International Review of Sport and Exercise Psychology*. doi: <https://doi.org/10.1080/1750984X.2020.1774915>
22. Thomas Zandonai and **Darias Holgado**. (2020). Doping in tennis, where we are and where we should be going? *Performance Enhancement & Health*. <https://doi.org/10.1016/j.peh.2020.100157>
23. **Darías Holgado**; Miguel A. Vadillo and Daniel Sanabria. (2019). ‘Brain-doping’, is it a real threat? *Frontiers in Physiology*. <https://doi.org/10.3389/fphys.2019.00483>
24. **Darías Holgado**; Thomas Zandonai and Daniel Sanabria. (2019). Comment on "Review of WADA Prohibited Substances: Limited Evidence for Performance-Enhancing Effects". *Sports Medicine*. <https://doi.org/10.1007/s40279-019-01062-4>
25. **Darías Holgado**; James Hopker; Daniel Sanabria and Mikel Zabala. (2017). Analgesics and Sport Performance: Beyond the Pain Modulating Effects. *PM&R*. 10.1016/j.pmrj.2017.07.068

Pre-prints

1. Juan-José Pérez-Díaz, **Darías Holgado**, Clara Alameda, Chiara Avancini, Luis Ciria, José-Antonio Salas-Montoro, & Daniel Sanabria. (2025). *Temporal Dynamics of Subjective Experience and Performance During a Dual-Task in Trained Cyclists*. *OSF*. https://osf.io/5b3uh_v1
2. Wolff, W., Hack, L., Halperin, I., **Holgado, D.**, Englert, C., Zech, A., Schüller, J., Bieleke, M., Shenhav, A., & Martarelli, C. (2025). *When effort fails: Instances and reasons for effort-performance decoupling*. *PsyArXiv*. https://doi.org/10.31234/osf.io/3uces_v1

3. Pérez-Díaz, J. J., **Holgado, D.**, Bekinschtein, T., & Sanabria, D. (2025). *A novel idiographic approach to map subjective experience in endurance sport*. PsyArXiv. https://osf.io/preprints/psyarxiv/g2bq4_v1/
4. **Darías, Holgado**, Víctor Martínez-Pérez, Elisa Martín-Arévalo, Tristan Bekinschtein, Miguel A. Fernández-del-Olmo, Daniel Sanabria, and Antonio Luque-Casado. (2024). «A Novel Approach to Study the Behavioral, Neural and Phenomenological Impact of Prefrontal HD-tDCS on Conflict Resolution». *PsyArXiv*,. <https://doi.org/10.31234/osf.io/cbwt3>.

Book chapter

In preparation: «*Chapter 20: Neurofeedback, Brain Stimulation, and Cognitive Enhancement*» in Cognitive Neuroscience in Sport and Athletic Performance. 2026. Routledge

Invited presentations

- Neuromodulation and athletic performance. **Academic Year:** Herbst 2024. **Faculty:** School of Psychology, Shenzhen University, China
- Fatigue Mentale, c'est quoi? Academic year: Fall 2022. PhD Graduate School of the University of Lausanne.
- Are we moving sport sciences forward? The need for effective and transparent practices. Academic year: Spring 2022. Faculty: Institute of Sports Science, University of Kyiv, Ukraine.

Mobility transfer activities

- Course: Module of the International Cycling Union's Level 3 Coach Course. "Fatigue and Recovery." October 2023; March 2024; May 2024
- Course: Module of the International Cycling Union's Level 3 Coach Course. "Self-paced exercise strategies." October 2023; March 2024; May 2024

Thesis-Panel

- Candidate: Svenja Wirtz. Dissertation: Mental Fatigue in Sport: From Individual Differences to Applied Practice (2026). University: La Trobe University.
- Candidate: Oliver, Liam. Dissertation: "The Time Course of Mental Fatigue and Its Effects on Cognition and Exercise Performance (2025)." University: Queensland University of Technology.

Conferences

Invited Symposium

1. **Holgado, D. et al.** (2026). « Exercise to preserve cognitive endurance performance? The role of self-paced exercise intensity at subjective and psychophysiological level » 31st Annual Conference ECSS Lausanne. Scheduled for July 2026. European college of Sport Science. Oral presentation.
2. **Holgado, D.**, Leubaz, L., Ruggeri, P., Borragán, G., Luque-Casado, A., Bekinschtein, T., Sanabria, D., & Place, N. (2025). *Is intense exercise bad for your mind? Subjective, behavioral and physiological responses to a cognitive effort to failure*. ISSP 16th World Congress of Sports Psychology. International Society of Sport Psychology. 2025. Hong Kong – Chair of « Integrating effort and performance dynamics in sport and exercise psychology: new theoretical and methodological insights ».

Presentations

3. **Holgado, D.**; Jolidon, L.; Borrigan, G.; Sanabria, D.; Place, N. « Individualized mental fatigue does not impact neuromuscular function and exercise performance ». 28th Annual Congress ECSS Paris 2023. European college of Sport Science. Oral presentation.
4. **Holgado, D.**; Zandonai, T.; Hopker, J.; Zabala, M.; Ciria, Luis F.; Sanabria, D. « Null effects of tDCS over the Left Prefrontal Cortex on Self-paced Exercise and EEG ». IV Science and Cycling Conference. UCI. 2018. France. Oral presentation.
5. **Holgado, D.**; Vadillo, Miguel A.; Sanabria, D. The Effects of tDCS on SportPerformance: a Meta-Analysis. XII. Kongress der Spanischen Gesellschaft für Experimentelle Psychologie (SEPEX). SEPEX. 2018. Spanien. Oral presentation
6. Zandonai, T.; **Holgado, D.**; Perakakis, P.; Luque, A.; Ciria, L.F.; Zabala, M. & Sanabria, D. Tramadol and cycling performance: a double-blind randomized placebo-controlled clinical trial European Brain Behavior Society 47th European Brain Behavior Society Meeting. 2017. Bilbao, Spain

Posters

7. **Holgado, D.**; Zandonai, T.; Hopker, J.; Zabala, M.; Ciria, L. F.; Sanabria, D. No effects of tDCS over the Left Prefrontal Cortex on Self-paced Exercise and EEG. XII. Kongress der Spanischen Gesellschaft für Experimentelle Psychologie (SEPEX). SEPEX. 2018. Madrid, Spanien.
8. **Holgado, D.**; Zandonai, T.; Hopker, J.; Zabala, M.; Ciria, Luis F.; Sanabria, D. Null effects of tDCS over the Left Prefrontal Cortex on Self-paced Exercise and EEG. I. Nationaler Kongress der Forscher im Bereich Ausbildung: Förderung der Interdisziplinarität (JIFFI). Universität Granada. 2018. Spanien.
9. Ciria, L.F.; Luque, A.; **Holgado, D.**; Sanabria, Daniel; Perakakis, Pandelis Tonic and transient oscillatory brain activity during and after acute exercise Sociedad Española de Psicología Experimental. XII. Kongress der Spanischen Gesellschaft für Experimentelle Psychologie (SEPEX). 2018. Madrid, Spanien.
10. Ciria, L.F.; Perakakis, P.; Luque, A.; **Holgado, D.** & Sanabria, D. Influence of acute physical exercise intensity Universidad de Granada Active brains for all: exercise, cognition and mental health 2017. Granada, Spanien.
11. **Holgado, D.**; Zandonai, T.; Zabala, M.; Sanabria, D. « Tramadol effects on physical performance and sustained attention during a 20-min indoor cycling Time-Trial: A clinical Trial ». ISSP 15th World Congress of Sports Psychology. International Society of Sport Psychology. 2017. Spanien.

2.2 Teaching

Cours

- Module: Statistics and Research Methods. Program: B.Sc. in Psychology. Academic Year: Fall 2025, Spring 2026. ECTS: 20. Department: Psychology. FernUni Switzerland
- Module: Behavioral Analysis and Motor Development. Program: B.Sc. in Sports Science. Academic Year: Spring 2023. ECTS: 3. Department: Sports Science. University of Granada, Spain.

- Module: Are We Moving Sports Sciences Forward? The Need for Effective and Transparent Practices. Master's Seminar. Academic Year: Spring 2022. Faculty: Institute of Sports Science, University of Lausanne, Switzerland.
- Module: Fundamentals of Sports IV: Cycling. Degree Program: B.Sc. in Sports Science. Academic Year: Spring 2018, 2019. ECTS: 6 Faculty: Sports Sciences. University of Granada, Spain.
- Module: Sports Specialization: Cycling. Degree Program: B.Sc. in Sports Sciences. Academic Year: Fall/Spring 2016/2017 & 2018/2019. ECTS: 6 Faculty: Sports Science. University of Granada, Spain.
- Module: Advanced Sports Course: Cycling. Degree: B.Sc. in Sports Science. Academic Year: 2017/2018. ECTS: 3. Faculty: Sports Science. University of Granada, Spain.

Professional Development in Education

- Intro to Distance Teaching. FernUni Schweiz,

Master students

- Priscilla Bärtschi: «Disentangling cognitive vs. physiological factors contributing to task failure: a pilot study». University of Lausanne, Switzerland. 01.07.2025.
- Alice Cailleux. «Does Mental fatigue Impact physical performance?» University of Lausanne, Switzerland. 28.06.2024
- Ludovic Leubaz. «Does running until exhaustion increase mental fatigue?» University of Lausanne, Switzerland. 02.02.2024.
- Léo Jolidon. «Distinguishing fatigue from fatigability: new perspectives from a ~~multidisciplinary~~ approach ». University of Lausanne, Switzerland. 02.02.2023
- Esther Troya. «Does mental fatigue impair physical performance? A replication study» University of Granada, Spain. 17.06.2019.

2.3 External Funding

- 1 Project:** « Acute aerobic exercise as a strategy to preserve mental performance and subjective well-being (*EXER-COGFIT*) ». **Funding Agency:** Spanish Ministry of Science. Type: Competitive. Role: Co-Principal Investigator. Principal Investigator: Antonio Luque Casado (Rey Juan Carlos University, Madrid). Duration: January 1, 2025–December 31, 2027. Amount 126.000 €. Contribution: Conceptualization, methodology, data collection, data processing, and formal analysis; editing. Abstract: The aim of this project is to investigate whether physical activity can effectively maintain cognitive performance and improve subjective experiences during sustained mental effort.
- 2 Project:** “Clinical trial on the effects of the combination of tramadol and paracetamol on physical, cognitive, and brain performance during cycling.” Funding agency: World Anti-Doping Agency (WADA). Type: Competitive. Role: Co-investigator. PI: Daniel Sanabria. (University of Granada). Duration: 05/08/2018–05/07/2020. Total funding: €112,000.00. Contribution: Design, methodology, data collection, data processing and formal analysis, writing (Article 9 in CV). Abstract: We tested the hypothesis that tramadol and acetaminophen improve athletic performance, but at the cost of reduced concentration.
- 3 Project:** “Putting the brain at work: Self-regulated acute aerobic exercise and executive control”. Funding agency: Spanish Ministry of Economy and Finance. Type: Competitive. Role: Co-

investigator. PI: Daniel Sanabria (University of Granada). Duration: 01/01/2017–31/12/2019. €54,500. Contribution: Design, methodology, data collection, data processing and formal analysis, writing (articles 12, 13, 14, 21, and 23 in CV). Abstract: The main objective of this project was to gain new insights into the reciprocal relationship between acute endurance training, the brain, and cognition, with a particular focus on executive cognitive function and its neural basis.

- 4 Project: “Tramadol and Sport: Effects on Physical and Sustained Attention Performance During Cycling Exercise.” World Anti-Doping Agency (WADA). Role: Co-Investigator. Principal Investigator (PI): Daniel Sanabria (University of Granada). Duration: April 15, 2016–April 15, 2017. Total funding: €75,000. Contribution: Design, methodology, data collection, data processing and formal analysis, writing (Article 5 in CV). Abstract: We investigated whether tramadol improves cycling performance and tested whether it affects cognitive functions (sustained attention) and brain function before, during, and after a cycling exercise.

Awards / Honors

- A prize of €1,000 from the grant program for publications in scientific journals and participation in activities organized by the Research Center for Mind, Brain, and Behavior at the University of Granada for the article: «Zapping the brain to enhance sport performance? Evidence from an umbrella review of the effect of transcranial direct current stimulation on physical performance. » *Neuroscience & Biobehavioral Reviews*.
<https://www.sciencedirect.com/science/article/pii/S0149763424002902?via%3Dihub>
- Competitive research grant from the Ministry of Education, Culture, and Sport. October 2015 to October 2019. 4-year contract
- Grant from the University of Kent: Canterbury, Kent, UK. Faculty of Sport and Exercise Science. July 10, 2015, to October 10, 2015. Total: £6,000
- Competitive research grant from the University of Granada. Faculty of Sports and Exercise Sciences. April 2015 to October 2015. Total: €2,000
- Master’s degree in Sports Science. May 17, 2015
- Competitive research grant from the Ministry of Universities. November 2013 to August 2014. Amount: €2,000

Languages

Language***	Listening	Reading	Writing	Speaking
French	C1	C1	C1	C1
German	C1	C1	B2/C1	B2/C1
Italian	C1	C1	B2	B2
English	C1	C1	C1	C1

*Self-assessment. **Native language: Spanish